第37手

Move 37



两步万分之一的棋，为什么令人惊叹？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Go is said to allow more configurations than the universe has atoms, a daunting computational challenge. In October 2015 DeepMind's AlphaGo reached a milestone, becoming the first program to beat a professional on a full board without a handicap. In March 2016 its opponent in a televised Seoul showdown was Lee Sedol, a South Korean strategist with 18 international titles.

In game two, on 10 March, AlphaGo's 37th move was a shoulder hit, a tactic placing a stone adjacent to an opponent's on the diagonal. On the grid's fifth line, it was hardly orthodox. Commentators found the unexpected placement bizarre; one took it for a mistake. Lee left the room, his pause for deliberation lasting over ten minutes.

DeepMind's diagnostic tools showed AlphaGo had put the odds of a human playing that stone at one in 10,000, a staggering rarity. AlphaGo had first learned to imitate strong amateurs online, then sharpened its play over tens of millions of simulated games against itself. Lee later acknowledged the stone's originality and beauty.

The fourth game, on 13 March, reversed the roles in neat symmetry. Lee's 78th move turned the tide, and many professionals heralded it as divine. DeepMind's team computed a professional's likelihood of finding it: about one in 10,000. Lee's only win was a consolation amid AlphaGo's overwhelming 4–1 victory.

In November 2019 Lee withdrew from the professional circuit, a farewell he tied to AI's unrivalled dominance. Calling AI an invincible entity, he said the summit was beyond him. Yet his one win came from a stone whose odds, by DeepMind's figures, were as slim as those of AlphaGo's audacious shoulder hit.

中文译文

据说，围棋可能出现的局面，比宇宙中的原子还多，这给计算带来了艰巨挑战。2015年10月，DeepMind的AlphaGo达成了一项里程碑：成为首个在全尺寸棋盘上、不让子的条件下击败职业棋手的程序。2016年3月，在首尔一场电视转播的对决中，它的对手是拥有18项国际冠军头衔的韩国围棋高手李世石。

3月10日的第二局，AlphaGo第37手下出一记肩冲，也就是在对方棋子的斜对角相邻处落子。这一手落在从棋盘边缘数起的第五线上，很不合常规。解说员觉得这个出人意料的落点十分怪异，其中一人还以为它下错了。李世石离开对局室，暂停下来思考了十多分钟。

DeepMind的诊断工具显示，AlphaGo曾估计，人类下出这一手的概率只有万分之一，罕见得令人惊叹。AlphaGo起初学习模仿网上实力强劲的业余棋手，随后通过数千万盘模拟自我对弈磨砺棋力。李世石后来也认可了这一手的独创性与美感。

3月13日的第四局，双方的角色巧妙地对调了。李世石第78手扭转局势，许多职业棋手将其誉为“神之一手”。DeepMind团队估算，职业棋手找到这一手的可能性约为万分之一。AlphaGo以4比1取得压倒性胜利，李世石唯一的一胜，则给他带来了一丝安慰。

2019年11月，李世石退出职业棋坛，并将这次告别与人工智能无可匹敌的优势联系起来。他称人工智能是不可战胜的存在，表示自己已无法企及那个巅峰。然而，带来他唯一胜局的那一手，按照DeepMind的数据，人类找到它的概率，和下出AlphaGo那记大胆肩冲的概率一样渺茫。

文中两次“万分之一”都是对人类选出某一步棋的概率估计，不是落下这步棋后的获胜概率。

核心词汇

先记这22项。连着下方的原文短语一起回想。

configuration n.
布局；局面
allow more configurations

daunting adj.
艰巨的；令人望而生畏的
a daunting computational challenge.

computational adj.
计算的
a daunting computational challenge.

milestone n.
里程碑
reached a milestone

handicap n.
让子；比赛中给对手的优势
without a handicap.

tactic n.
战术；招数
a tactic placing a stone

orthodox adj.
正统的；合乎常规的
hardly orthodox.

bizarre adj.
怪异的；异乎寻常的
found the unexpected placement bizarre

deliberation n.
审慎思考；斟酌
pause for deliberation

diagnostic adj.
诊断的；用于分析问题的
DeepMind's diagnostic tools

odds n.
概率；可能性
the odds of a human playing that stone

imitate v.
模仿
imitate strong amateurs online

simulated adj.
模拟的
simulated games against itself.

originality n.
独创性
the stone's originality and beauty.

symmetry n.
对称；相互呼应
neat symmetry.

turn the tide phr.
扭转局势
Lee's 78th move turned the tide

likelihood n.
可能性；概率
a professional's likelihood

consolation n.
安慰
Lee's only win was a consolation

overwhelming adj.
压倒性的
AlphaGo's overwhelming 4–1 victory.

unrivalled adj.
无可匹敌的
AI's unrivalled dominance.

invincible adj.
不可战胜的
an invincible entity

audacious adj.
大胆的；敢于冒险的
AlphaGo's audacious shoulder hit.

文中词汇与搭配

补充词汇共29项。保留每个表达在本文中的意思。

atom n.
原子

opponent n.
对手

televised adj.
电视转播的

showdown n.
对决；决战

strategist n.
战略家；善于布局的高手

adjacent to phr.
与……相邻的

diagonal n.
斜线；对角线方向

grid n.
网格；此处指棋盘格线

unexpected adj.
出人意料的

placement n.
放置；此处指落子位置

pause n.
暂停

staggering adj.
惊人的；令人震惊的

rarity n.
罕见；罕见之事

amateur n.
业余爱好者

sharpen v.
磨砺；提高（技艺）

acknowledge v.
承认；认可

reverse v.
颠倒；互换

neat adj.
巧妙的；恰到好处的

herald v.
称赞；誉为

as prep.
与herald连用，表示誉为

divine adj.
神妙的；此处指神之一手

compute v.
计算；估算

withdraw from phr. v.
退出；撤离

circuit n.
职业巡回赛事；职业赛场

farewell n.
告别

dominance n.
主导地位；优势

entity n.
实体；独立存在

summit n.
顶峰；巅峰

slim adj.
微小的；渺茫的（概率）

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In October 2015 DeepMind's AlphaGo reached a _____, becoming the first program to beat a professional on a full board without a handicap.

02 Commentators found the unexpected placement _____; one took it for a mistake.

03 Lee later acknowledged the stone's _____ and beauty.

04 Lee's only win was a _____ amid AlphaGo's overwhelming 4–1 victory.

文章主线

- 训练 围棋程序AlphaGo先模仿高水平业余棋手，再经数千万盘自我对弈提升，2016年迎战李世石。
- 奇手 第二局第37手落在从棋盘边缘数的第五条线上，违背常规，起初被疑为失误，后来获得李世石赞赏。
- 回应 第四局，李世石第78手扭转局势，被誉为神之一手，带来他在五局中唯一的胜利。
- 赛果 AlphaGo最终4比1获胜；李世石2019年退役时称AI不可战胜，自己已难登顶。
- 对照 AlphaGo估计人类下出第37手的概率约万分之一；团队对职业棋手下出第78手的估计相同，双方都有罕见创意。

参考答案　1 milestone 2 bizarre 3 originality 4 consolation